13 September 1968
SUPERSEDING
PA-PD-2545
7 March 1964

MILITARY SPECIFICATION

PROPELLANT M1 TYPE II (MASTER LOT) FOR USE IN CHARGE: XM72

1. SCOPE

1.1 This specification covers special requirements not included on the drawings, and provides the quality assurance provisions for the fabrication, assembling, verification of charge weight for XM72 Charge Zones 1 and 2, and the preparation for delivery for one type of propellant used in charge XM-72 and designated as Propellant M1 Type II.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-14766(MU) - Charge, Propellant: XM72 for 155mm
MIL-S-14770(MU) - Shell, Test, Velocity, 155MM, T5119 Assembly

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defects
MIL-STD-652 - Propellants, Solid for Cannons Requirements and Packing
MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1168 - Lot Numbering of Ammunition

PUBLICATIONS

ORDNANCE CORPS

Ordnance Proof Manual

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications.

3.2 Assembly. The propellant assembled in loading charges shall comply with MIL-C-14766(MU).

3.3 First article testing. Requirements for the submission of first article samples shall be as specified in the contract (see 4.2).

3.4 Propellant, composition and physical tests. The propellant shall comply with the requirements of MIL-STD-652 for M1 Propellant Type II, when tested as specified in 4.4.2.

3.5 Ballistics. The propellant shall be assessable to impart zone velocities, pressures, and standard deviation of velocities (see 6.2) with limits specified in Table I.

Table I
Ballistic requirements

Zone	Velocity	Min. Pressure Absolute	Max. Pressure Absolute	Max. Std. Deviation of velocity
1	1020	19200	22800	5.0
2	1230	30600	34600	5.0

3.6 Workmanship. The best commercial practices shall be used in the formulation of the propellant furnished under this specification, and in all other applicable documents. The propellant and its standard ingredients shall be protected from action of direct sunlight and acid fumes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

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4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of one or more batches of M1 Propellant produced by one manufacturer, in accordance with the same specification or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of M1 Propellant which has been subjected to the same unit chemical or physical mixing process intended to make the final product homogenous. Lot numbers shall be selected in accordance with MIL-STD-1168.

4.3.2 Inspection testing.

4.3.2.1 Propellant testing. The contractor shall perform all the tests required in MIL-STD-652.

4.3.3 Testing. The contractor shall perform all the tests specified in this paragraph except test a. Tests do not have to be performed in the order specified. Test classification critical, major and minor is defined in MIL-STD-105.

Test	Test classification
a. Ballistic	Major
b. Composition and physical	Major
c. Surveillance	Major

Of these tests a, b, and c are considered destructive and samples so tested shall not be returned to the lot. The destructive samples are over and above the deliverable quantity of propellant required by the contract.

4.3.3.1 Ballistic tests. (Aberdeen Proving Ground) (see 3.5). Propellant shall be randomly selected from the lot as specified in Table II and tested in accordance with 4.4.1. Any charge which fails to comply with 3.5 shall reject the lot.

Table II
Propellant required for samples

Propellant	Zone	Amount in Pounds
M1 Type II	1	100
M1 Type II	2	150

4.3.3.2 Composition and Physical tests. (see 3.4). A random sample selected from each lot of propellant shall be tested in accordance with paragraph 4.4.2. Any sample on any required test failing to comply with 3.4 shall reject the lot.

4.3.3.3 Surveillance Tests. A five pound sample of propellant shall be selected at random from each lot of propellant and sent to Commanding Officer, Picatinny Arsenal, Dover, N. J. All samples shall be packed in airtight containers and shall be marked to show the propellant designation, lot number manufacturer, date of sampling, contract number, number of pounds in lot. The expense of surveillance testing shall be borne by the

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Government. This sample will be used to detect deterioration in the lot resulting from storage so shall not be used for acceptance or rejection of the original lot of powder.

4.4 Test methods and procedures. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1. Ballistic test. Propellant shall be randomly selected from the lot as specified in Table II and sent to Commanding Officer, Picatinny Arsenal, Dover, New Jersey. The sample shall be packed in air tight containers and shall be marked to show propellant designation, lot number, manufacturer, date of sampling, contract number, number of pounds in the lot. Thirty-two charges shall be fabricated as specified in MIL-C-14766(MU). These charges shall be ballistically tested as specified in Table III with charges conditioned to + 70 ± 5 degrees fahrenheit in accordance with the applicable provisions of the Ordnance Proof Manual, unless otherwise specified herein, at Aberdeen Proving Ground, Aberdeen, Maryland. The T-5119 test shell shall be used as the test round in this test. The charges shall be conditioned for a minimum of 16 hours at 70 ± 5 degrees fahrenheit. The charges shall be removed from the chamber and fired within 10 minutes. The following data shall be recorded for each charge fired.

1. Velocity
2. Pressure
3. Date fired

Calculate and record for each 7 round series fired:

1. Mean velocity
2. Standard deviation (see paragraph 6.2)

Table III

8 rounds with existing propellant charge Zone I
8 rounds with existing propellant charge Zone II
8 rounds with new propellant Zone I
8 rounds with new propellant Zone II

32 rounds total.

4.4.2 Composition and Physical Tests (see 3.4). The sample shall be tested in accordance with MIL-STD-652.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking of propellant. Preservation, packaging, packing and marking of the propellant shall be in accordance with MIL-STD-652.

5.2 Preservation, packaging, packing and marking of charges shall be in accordance with MIL-C-14766(MU).

5.3 Data cards. Data cards shall be prepared in accordance with MIL-STD-1167.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Process - Details of manufacturing process and equipment used by the contractor shall be submitted to the procuring activity in writing prior to the commencement of manufacture. Any deviation from this manufacturing process shall be submitted to the procuring activity prior to being put into effect.

6.2 Standard deviation. The standard deviation shall be calculated with (N-1) as the divisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, example B1.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1145-A003

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